PharmaPoint:

Sales Productivity & Performance Analysis

Distinguishing Business Scale from True Performance



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PostgreSQL • BigQuery • Looker Studio

Business Context

Revenue differences are visible, but their drivers are not obvious

Two Questions Quietly Shape PharmaPoint's Performance Measurement

Problem 1
Regional Interpretation

High-revenue provinces may simply have more branches. Total sales alone can misclassify network scale as operational performance.

Problem 2
Product Interpretation

High-revenue products or categories may reflect higher prices or more SKUs, not stronger transaction frequency.

Key Business Questions

1. Is sales performance changing over time?
2. Are high-revenue regions truly more productive?
3. What drives category and product revenue?
4. Do discount and rating explain performance?
5. Can inventory support stock-to-sales analysis?

Case objective

Build a performance framework that separates business scale from true productivity, then turn the findings into an interactive dashboard.

Data & Tools

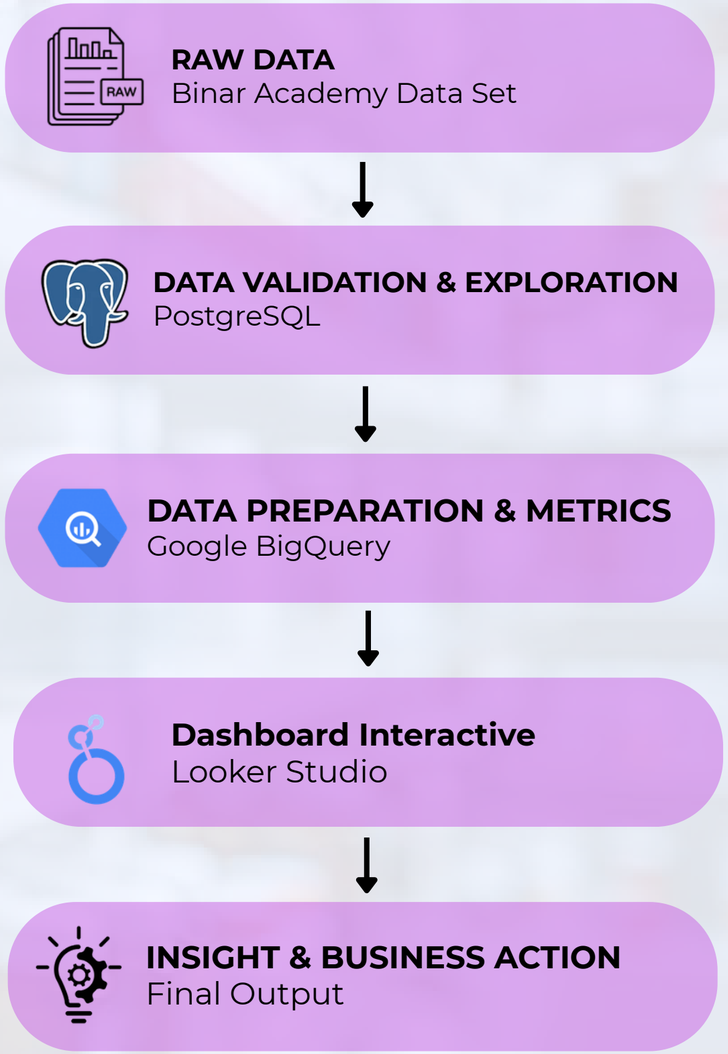
Built from four raw tables and transformed into one dashboard-ready analytical view

Data Set Information

Table Name	Descriptions	Column	Scope
final_transaction	Transaction-level sales records including date, branch, customer name, product, price, discount, and rating	8	672,458 rows
kantor_cabang	Branch master data containing branch category, branch name, city, province, and branch rating	6	1,725 branches
product	Product master data containing product name, category, and price	4	150 SKUs
inventory	Inventory observations containing branch-product stock records	5	1,035,000 rows

Date range: 2020-01-01 to 2023-12-30.
Inventory was evaluated separately due to missing timestamp.

Analysis Flow & Tools



Part 0:
Data Foundation

Clean first, Interpret later.



Data Validation & Preparation

Four Source Tables Validated, Then Integrated Into One Analytical View.

Table Name	Validation & Preparation Details
final_transaction	– Checked key fields for missing values → 0 NULLs found – Validated branch_id and product_id against master tables → 0 orphan IDs – Checked price, discount, and transaction rating ranges → 0 invalid values – Compared transaction prices with product master → 0 price mismatches
kantor_cabang	– Verified branch_id uniqueness → 1,725 unique branches – Checked branch name, category, city, and province fields for completeness – Validated branch rating range → all values between 3.9–5.0 – Confirmed branch IDs are consistently referenced in transaction and inventory tables
product	– Verified product_id uniqueness → 150 unique SKUs – Checked product name, category, and reference price fields – Compared product names with inventory records → 0 mismatches – Compared product prices with transaction records → 0 inconsistencies
inventory	– Checked stock values → 0 negative stock records – Validated branch and product references → 0 orphan IDs – Identified 254,004 unique branch-product pairs – Found 92.5% of pairs have repeated observations, but no timestamp is available to determine chronology
vw_sales_enriched	– Joined transaction data with branch and product attributes – Preserved 672,458 transaction rows after enrichment – Added calculated net_sales = price × (1 − discount_percentage) – Inventory excluded to avoid one-to-many duplication of transaction records

Sales, branch, and product data passed the required integrity checks. Inventory was retained as a separate reliability assessment because repeated observations cannot be ordered without timestamps.

Feature Engineering & Metric Framework

Building metrics that separate business size from performance

ANALYTICAL LAYER

final_transaction

+

kantor_cabang

+

product

=

vw_sales_enriched

Delivered field:
net_sales = price * (1-discount_percentage)

Grain:
1 row = 1 transaction record.

672,458 rows preserved
after enrichment

PERFORMANCE FRAMEWORK

Scale Matrics
Total Net Sales
Total Transactions
Total Branches
Total SKU

Productivity Metrics
Net Sales / Branch
Transactions / Branch
Net Sales / SKU
Transactions / SKU

ANALYTICAL OUTPUTS

Output	For
Sales Trend	Time Performance
Provincy & City Matrics	Geographic Performance
Category & Product Matrics	Product Performance
Discount & Rating Matrics	Commercial Analysis
Inventory Grain Audit	Data Reliability

Scale ≠ Performance

Analytical principle

Every scale metric is paired with a normalized productivity metric so business size is not mistaken for superior performance.

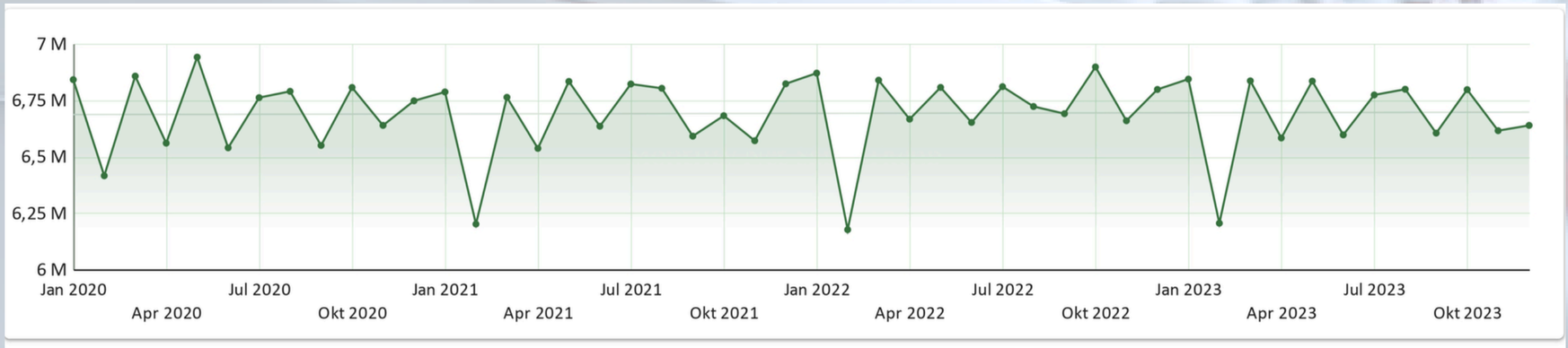
Part 1: Regional Performance

Are high-revenue regions truly more productive or simply larger?



Finding 1 — Sales Stayed Flat for Four Years

Time is not the primary performance driver in the dataset.



Annual Summary	
2020	**Rp 80.44B**
2021	**Rp 80.04B**
2022	**Rp 80. 58B**
2023	**Rp 80.12B**

Conclusion: Time is not the main performance driver.

Evidence

Both annual and monthly views show a highly stable sales pattern. Even after validating monthly totals with sales per active day, **performance continued to return to a similar baseline.**

Why it matters

Time is not the main performance driver. The analysis should shift from searching for growth or seasonality toward **explaining differences in business scale, regional productivity, and product structure.**

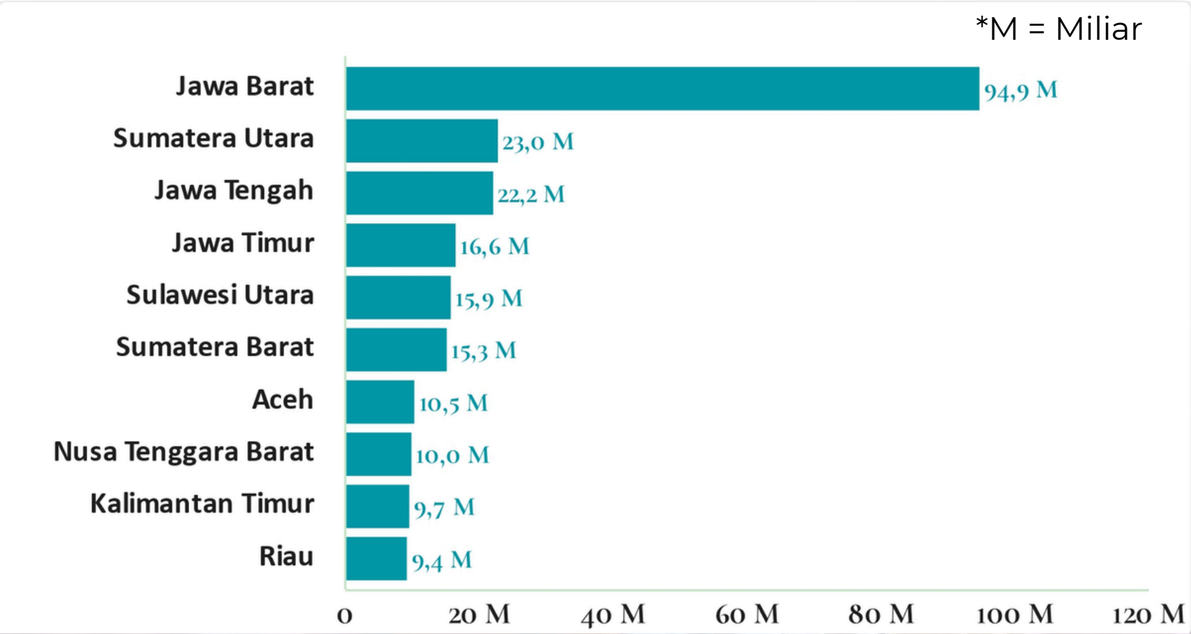
Key Evidence

1. Annual YoY movement stayed within **±1%**
2. Monthly range remained around **Rp6.2B – Rp6.9B**
3. Sales per active day also remained **stable**

Finding 2 — High-Revenue Regions Are Larger, Not Meaningfully More Productive

Jawa Barat dominates total revenue, but not branch productivity

Total Net Sales by Province



Scale Evidence



Jawa Barat

510 Branches

RP94.9B Net Sales

Productivity Context

Company benchmark	**Rp186.2M / branch**
Jawa Barat net sales / branch	**Rp186.0M**
Gap vs company average	**-0.09%**

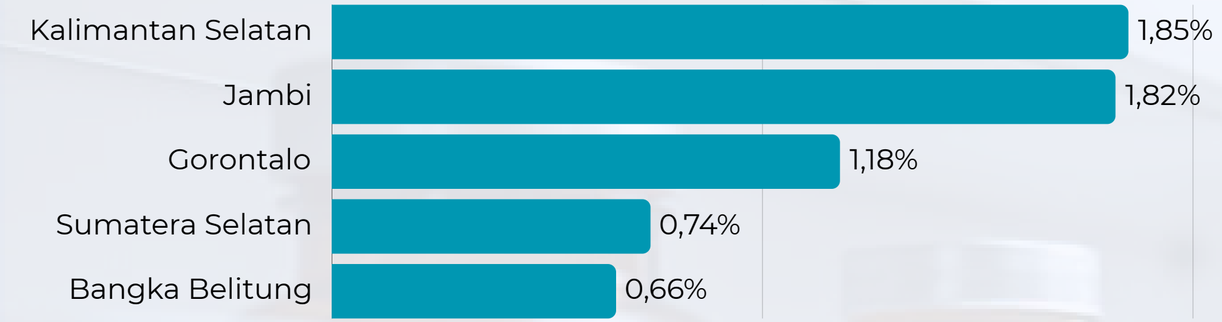
Why it Matters?

This province leads because it operates at a much larger branch network, not because each branch performs dramatically better. Scale and productivity should be interpreted separately

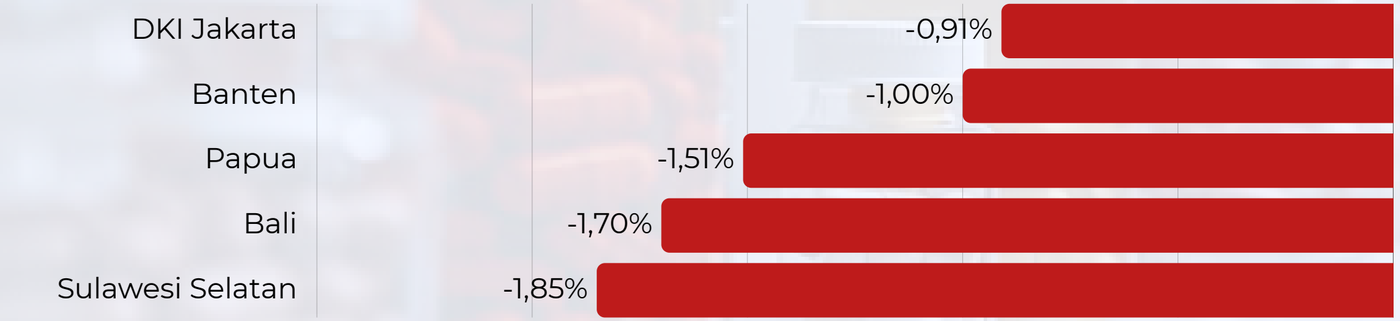
Finding 3 — Regional Productivity Differs by Only a Few Percent

Ranking differences exist, but the actual productivity gap is small

TOP PROVINCES VS AVG



BOTTOM PROVINCES VS AVG



***Company benchmark =** Rp186.2M / branch*

Evidence

Province-level branch productivity ranges from **+1.85% to -1.85%** against the company benchmark. City-level variation is also limited.

Why it Matters

A ranking can look dramatic while the actual gap is operationally small. Large strategic moves need stronger evidence than small productivity differences.

Key takeaway: Total spread ≈ 3.7 percentage points.

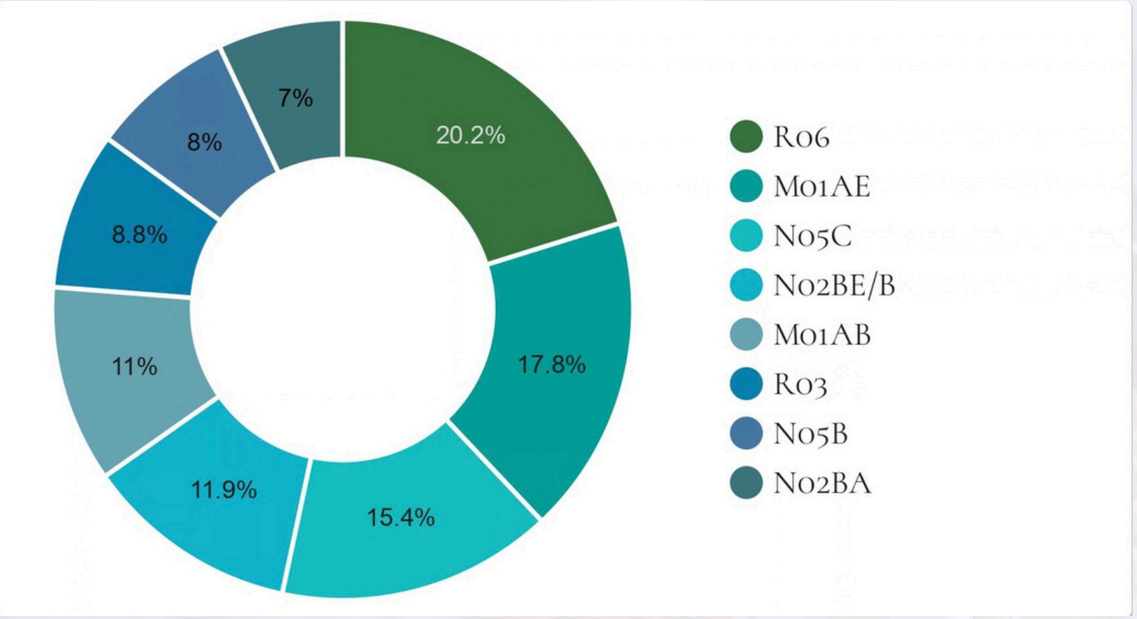
Part 2: Product & Commercial Drivers

What actually explains revenue differences across products and categories?



Finding 4 — Category Revenue Reflects Assortment Size and Price Mix

Transaction frequency per SKU remains nearly identical across categories



Category Evidence

*Tx = Transactions

Category	SKU	Tx / SKU	Net Sales
R06	30	4,493	Rp64.9B
M01AE	23	4,491	Rp57.1B
N05C	27	4,478	Rp49.3B
M01AB	14	4,471	Rp35.2B

Tx / SKU remains tightly clustered **(~4,461–4,497),** while **Avg Sales / Transaction** varies more materially **(~Rp408K–Rp563K).**

Why It Matters

Because **SKU-level transaction frequency is almost uniform,** category revenue differences are better **explained by assortment size and price/value mix** than by materially higher frequency.

Finding 5 — Product Revenue Is Almost Perfectly Explained by Price

Product Revenue Is Overwhelmingly Price-Driven” atau “Product Revenue Closely Tracks Price.

0.9996

Price ↔ Net Sales

-0.1604

Transactions ↔ Net Sales

0.0869

Discount ↔ Net Sales

Product-level evidence	Value
Product transactions min	4,330
Product transactions max	4,608
Price range	Rp 2,100 – Rp 997,500
Highest revenue product	KF953

Interpretation



High revenue should not be read as “highest demand.” In this dataset, product revenue is overwhelmingly price-driven.

Portfolio caution



Use “highest-revenue product,” not “best-selling product,” because quantity sold is not available and transaction frequency is highly uniform.

Finding 6 — Discount & Rating Are Weak Performance Signals

Commercial indicators show little linear association with sales performance in the available data.

-0.131

Discount ↔ Transactions

-0.006

Rating ↔ Net Sales

-0.015

Rating ↔ Transactions

+0.014

Rating ↔ Avg Transaction

What this means

No meaningful linear relationship was observed between average discount and transaction frequency, or between branch rating and sales performance.

This does not prove that discount or rating has no causal effect.

This is not evidence that discount is ineffective. The dataset lacks experimental design, exposure, margin, and control groups.

Discount correlations are calculated at product level; rating correlations at branch level.

Inventory Reliability — 1.03M Records, But No Reliable Timeline

The issue is not stock value — it is missing chronology

Inventory grain check	Result
Inventory records	1,035,000
Unique branch-product pairs	254,004
Repeated pairs	234,978
Repeated share	92.5%
Average records / pair	4.07
Maximum records / pair	17
Timestamp available?	**No**

Why it matters

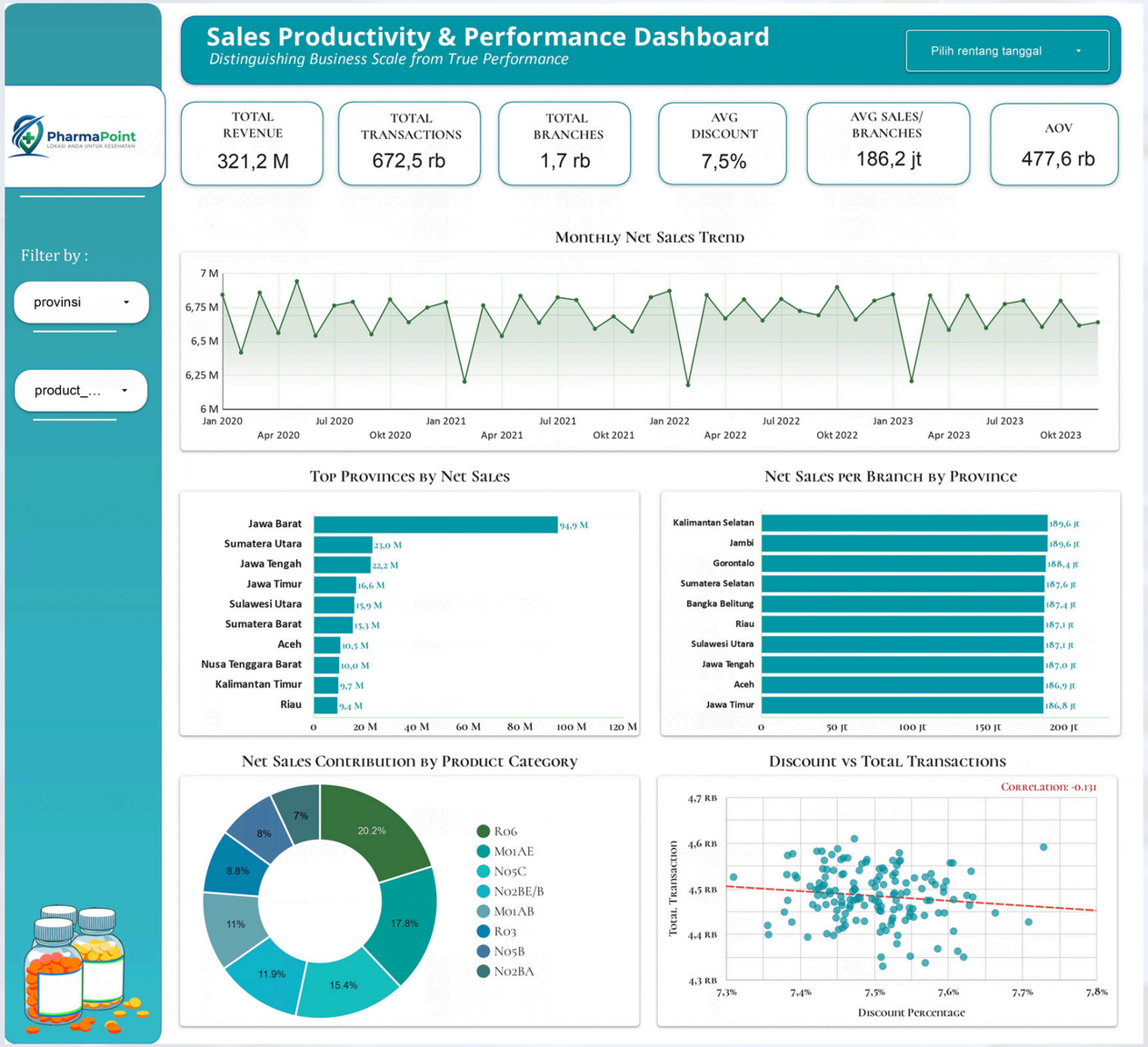
Without timestamps, current/latest stock and stock movement cannot be determined. Coverage, turnover, and stock-to-sales analysis also require additional quantity/cost context.

Data recommendation

Target grain:
1 row = 1 branch + 1 product + 1 snapshot timestamp.

From Findings to a Living Tool

Core performance findings are consolidated into one interactive dashboard



Top Row — Executive KPI
Revenue, transactions, scale, discount, productivity, and AOV.

Upper Section — Sales Trend
Confirms stable monthly net sales across 2020–2023.

Middle — Scale vs Productivity
Separates regional revenue contribution from Net Sales per Branch.

Bottom — Product & Commercial Signal
Category contribution and discount vs transaction frequency.

Note: This dashboard is a single analytical narrative, not merely a monitoring interface.

Dashboard Overview

Link

Part 3:
What To Do

Three pillars, based only on evidence the dataset supports.



Pillar 1 — Fix Performance Measurement

Adopt scale-adjusted regional KPIs before making regional decisions

Measurement choice	Recommendation
Use together	**Total Net Sales + Net Sales / Branch**
Use together	**Total Transactions + Transactions / Branch**
Benchmark against	**Company Net Sales / Branch average**
Avoid	**Ranking-only decisions**

CORE IDEA ⟶ Scale ≠ Performance

Interpretation rule

Treat Total Net Sales as market scale and Net Sales / Branch as productivity. A region is not **"better"** simply because it is larger.

Decision caution

Do not make expansion, closure, or resource-allocation decisions from small rank differences **without additional operational evidence.**

Pillar 2 — Improve Product Performance Reporting

Separate revenue contribution from transaction frequency and price mix

Product lens	Metric
Revenue contribution	**Net Sales**
Assortment scale	**SKU Count**
Transaction frequency	**Transactions / SKU**
Transaction value	**Avg Sales / Transaction**
Price mix	**Average Product Price**

Metric ⟶ **Revenue ≠ Demand**



Interpretation rule

A high-revenue product or category should not automatically be treated as high-demand. Revenue must be read together with price and transaction frequency.



Labeling rule

Avoid using “best-selling” unless quantity sold is available. Use “highest-revenue” or “highest transaction frequency” depending on metric.

Pillar 3 — Make Inventory Temporally Usable

Stock analysis requires a timestamped inventory grain

Current grain
branch_id
product_id
opname_stock

Target grain
branch_id
product_id
snapshot_datetime
opname_stock

What a timestamp enables

Every branch-product stock observation can be chronologically ordered. Only then should the business evaluate current stock, movement, coverage, or stock-to-sales alignment.

Do not do yet

Do not calculate overstock, understock, turnover, or replenishment priority from the current inventory table.

CORE IDEA ⟶ Target grain: 1 row = 1 branch + 1 product + 1 snapshot timestamp

Limitations & Methodology Notes

Known constraints define the scope of the recommendations

Stable metrics

Several synthetic-data metrics are unusually stable; magnitude-based findings should not be generalized directly to real pharmacy operations.

No units sold

Revenue and transaction count cannot be translated into units sold.

No profitability lens

Profitability and margin cannot be calculated because cost data is unavailable.

Customer field limits

Retention or loyalty analysis is not reliable from the available customer field.

Inventory chronology missing

Stock chronology, coverage, and stock-to-sales analysis cannot be reliably determined without timestamps.

No causal inference

Discount and rating relationships do not prove causal impact.

KEY TAKEAWAY ⟶ Interpret findings within the scope of the available data.

Conclusion

The strongest story is not a dramatic sales problem — it is a measurement problem

1

Revenue concentration is mostly explained by **network scale.** Pair total revenue with productivity metrics.

2

Category and product revenue are mostly shaped by assortment size and price mix, not transaction frequency gaps.

3

Discount and branch rating show little linear association with performance in the current dataset

4

Inventory needs snapshot timestamps before stock-performance decisions can be supported reliably.

5

Dashboard, SQL queries, technical documentation, and GitHub case study will consolidate the end-to-end workflow.

EXECUTIVE TAKEAWAY

Separate scale from productivity before making business decisions.

Project Artifacts & Codebase

Github:
https://github.com/rizalajiwiguna/Pharmacy-Project

Looker Studio:
https://datastudio.google.com/reporting/26553675-e866-47cc-8025-376b317c85aa

PosgreSQL & BigQuery:
https://github.com/rizalajiwiguna/Pharmacy-Project/blob/main/queries.sql

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